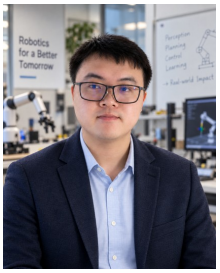
Hong Wei Soon

Robotic Sensing and Perception Researcher

Birmingham, United Kingdom
marssoonhongwei@gmail.com
LinkedIn | ORCID 0009-0006-3612-2773 | FieldSense project



Research profile

PhD researcher studying terrestrial robotic inspection and adaptive sensing. Current work combines ROS 2 and Unitree A2 data collection, LiDAR and visual SLAM workflows, environmental sensing and gas-dispersion simulation. Research interests include measurement reliability, gas-source localization and perception for infrastructure inspection.

Education

PhD in Civil Engineering - in progress Feb 2025 - present
University of Birmingham, United Kingdom

Research: terrestrial robotic inspection, gas sensing and adaptive measurement. Supervisors: Wonjun Cha, Mehran Eskandari Torbaghan and Amir Hajiyavand.

MSc Mechatronics with Merit 2024
University of Glasgow, United Kingdom

Selected coursework: advanced control, autonomous vehicle guidance, robotics and digital signal processing.

BEng Honours Mechatronics Engineering 2019 - 2022
Universiti Tunku Abdul Rahman, Malaysia

Doctoral research

PhD Researcher Feb 2025 - present
University of Birmingham, Department of Civil Engineering

Unitree A2 perception and inspection

- Run ROS 2 data-acquisition workflows on a Unitree A2, collecting real LiDAR and camera observations for SLAM and infrastructure-inspection research.
- Develop LiDAR and visual SLAM workflows linking robot pose and scene mapping to inspection measurements.
- Define coordinate-frame, timestamp and sensor-position requirements for connecting LiDAR and camera maps with gas measurements; document the interface for collaborative integration.

Adaptive sampling and environmental sensing

- Conduct mobile humidity-sensing experiments comparing continuous measurements with timed stationary sampling under fan and no-fan conditions.
- Analyze sensor response and measurement variation to guide sampling intervals, dwell time and robot motion; develop adaptive inspection for gas-sensing applications as ongoing research.
- Investigate how sampling height, robot pose and sensor response affect the spatial assignment of measurements, with physical humidity tests and gas-dispersion simulation treated as separate evidence.

Awards and languages

School Studentship Scholarship, University of Birmingham: tuition fees and stipend for 3.5 years of doctoral research. President's List, Universiti Tunku Abdul Rahman, 2019.

English, Mandarin and Malay; basic German.

Research software and outputs

Gas dispersion and sensor-response simulation

- Developed an interactive JavaScript tool comparing four formulations: finite-volume advection-diffusion, Gaussian puffs, random-walk particles and a steady Gaussian reference.
- Added configurable wind, obstacles, release height, vertical mixing and sensor response/recovery, with CSV and settings exports for reproducible comparisons.
- Implemented numerical sanity checks and documented equations, assumptions and a proposed sensor-to-map interface. The tool supports exploratory analysis; physical validation remains future work.

Open gas sensing field lab (sign-in required)

FieldSense sensor selection website

- Developed a web tool for preliminary comparison of sensing technologies against inspection requirements and operating conditions.
- Organized sensing principles, capabilities, limitations and integration requirements to support candidate screening before device testing.

Review manuscript in preparation

Critically reviews terrestrial robot platforms, sensing technologies and autonomy for infrastructure inspection. Develops a platform-configuration framework and adaptive sampling guidance linking inspection requirements, sensor dynamics and operating conditions.

Object recognition and detection - personal project (2026)

- Built a browser-based playground for recognising objects from webcam images and uploaded photos, with pretrained detection models and optional cloud vision.
- Added image-region selection and timestamped recognition feedback; documented model limitations.

Technical skills

Robotics and perception: ROS 2; Unitree A2; LiDAR and camera data acquisition; LiDAR and visual SLAM workflows; Unitree Go2 demonstrations.

Sensing and methods: Gas and environmental sensing; sensor response and recovery; sampling design; experimental comparisons; gas-dispersion simulation.

Software: C++, JavaScript, foundational Python, MATLAB, Git and GitHub; LaTeX and Overleaf.

Hardware and design: Arduino, SolidWorks and AutoCAD; academic training in Epson robots, Omron collaborative robots and PLCs; Node-RED.

Manufacturing experience

Industrial Trainee Oct - Dec 2021

SAM Meerkat (M) Sdn Bhd, Penang, Malaysia

- Prepared and revised assembly work instructions, bills of materials and checklists for semiconductor equipment; maintained engineering-review records in eQMS.
- Assisted with torque-driver calibration, O-ring inspection and filter, regulator and lubricator checks under engineering supervision.
- Picked, scanned and issued components to production; supported stock counts, packing, dispatch preparation and trainee handover.

Engineering communication and outreach

- Demonstrated industrial robotics and the Unitree Go2 at a RAS West Midlands and Robotics Adoption in the Midlands Programme event; supported school outreach at Birmingham.
- Prepared No Dig Live presentation materials on robot dogs for construction and infrastructure inspection, covering the A2 test field, operator camera views and intended LiDAR mapping.